Ambareesh Shyam Sundar

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Education

Georgia Institute of Technology • GPA: 4.0 / 4.0 **Atlanta, GA**
B.S. in Computer Science; B.S. in Mathematics *August 2023 - December 2026*
Selected coursework: Operating Systems, Processor Design, Computer Architecture, Advanced Algorithms, Natural Language Processing, Computer Networking, Probability and Statistics

Skills and Awards

Languages: Python, C, C++, Rust, Go, Javascript, Typescript, HTML/CSS, React
Technologies: Linux, Git, Github, Docker, AWS, MySQL, Postgres
Languages: English, Tamil, Chinese (Mandarin)
Awards: Putnam score of 23; 5x Georgia Tech Faculty Honors; College of Sciences Multidisciplinary Award Nominee

Experience

GT/IBM Summer Internship Program **May 2024 – August 2024**
Software Engineering Intern *PyTorch, Python, C++*
- Contributed bug fixes and documentation to PyTorch repositories during IBM mentorship program.
- Diagnosed and resolved defects in TorchDynamo JIT compilation pathways, improving correctness and stability.

Active Sensing in Masked Autoencoders **January 2024 – present**
Research Assistant, Georgia Tech Computational Neuroscience Group *PyTorch, computer vision*
with Dr. Hannah Choi and Abdo Sharaf
- Designed and built a two-stage vision pipeline on top of Masked Autoencoders (MAE) that performs uncertainty-aware patch selection prior to encoding, reducing redundant computation and improving downstream performance.
- Validated the system on ImageNet by replacing random patch masking with intelligent selection, demonstrating up to a 4.5% improvement in classification accuracy with minimal additional inference overhead.

Projects

Foxtail • ambareesh1510/foxtail **November 2025 – present**
Unix-like operating system kernel for 32-bit x86. *C, bare-metal programming, QEMU, GDB*
- Designed and implemented a fully functional Unix-like x86 operating system kernel and user-space from scratch, supporting preemptive multitasking, real-time graphics, and execution on real hardware.
- Developed a flexible process management interface, a custom filesystem with support for files, directories, and symbolic links, and inter-process communication via pipes.
- Built a complete memory management system with paging, virtual memory, demand allocation, and strong isolation/safety guarantees enforced by hardware protection rings.

Open Aim Trainer • ambareesh1510/openaimtrainer-frontend **July 2025 – August 2025**
Online, open source FPS trainer. *Oracle Cloud, MySQL, Docker, REST API, Go, C*
- Engineered a high-performance, cross-platform aim training platform with backend deployed on Oracle Cloud using Docker, Nginx, SSL security, and REST APIs.
- Built simulation engine capable of handling 100k concurrent active entities while maintaining stable frame performance.

Dungeon Deja Vu • ambareesh1510/dungeon-deja-vu **July 2024**
Puzzle-platformer for Bevy game jam. *Rust, Bevy (ECS)*
- Developed puzzle-platformer in one week for Bevy Game Jam 5; awarded 3rd place out of 80 participants.
- Used Rust and Bevy ECS to enable fast iteration and robust game logic, creating over 2 hours worth of gameplay.